

Airflow Angle and Wind Estimation Using GPS/INS Navigation Data and Airspeed

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Abstract: The aircraft autopilot needs the feedback of airflow angles, that is, angle-of attack and sideslip angle to control the aircraft. The airflow angles measured by a mechanical vane are not accurate and very noisy for low airspeed. Moreover, the vane has a mechanical failure risk. This paper describes the extended Kalman filter based method to estimate the airflow angles and three-dimensional wind speed under constant wind condition. In addition, it can correct the scaling error of the airspeed of an aircraft. It uses the airspeed measurements, constant wind condition and the sideslip angle computed from GPS/INS navigation data and stability and control derivatives estimated from flight data. Simulation results show that the proposed method works well in various conditions. It is expected that estimated airflow angles will be able to replace the airflow angles measured by a primary system. Estimated wind speed can also be used to reconstruct the GPS/INS navigation system by correcting the airspeed in the case of GPS failure.

Keywords: Airflow Angles, Angle-of-Attack, Angle-of-Sideslip, Wind Estimation, Airspeed, GPS/INS Navigation, Kalman Filter

1. INTRODUCTION

Most of the aircraft autopilots need airflow angles, that is, angle-of-attack and side-slip angle to control the aircraft. Airflow angles can be measured easily by mechanical vanes. However, measuring airflow angles by these vanes should have the backup system because it has a mechanical failure risk. The wind information is generally obtained from ground and airspeed if airflow angles are measured accurately. Wind information can be used to reconstruct GPS/INS navigation system by correcting the airspeed of an aircraft in the case of GPS failure. Wind estimation is also one of many applications of the UAV.

There are many methods to estimate airflow angles and wind speed from navigation data, not to measure directly. Myschik et al. used the complementary filter configuration to combine the airflow angle from aerodynamic model and navigation data [1]. Hao Long computed airflow angles by estimating wind speed from navigation data and aerodynamic model. Kevin A. Wise developed the extended Kalman filter method based on decoupled longitudinal and lateral aerodynamic model and GPS/INS navigation data [3]. Reference [4] estimated wind speed from GPS data and airspeed measurements under the maneuver that the heading of an aircraft changes.

This paper proposes the extended Kalman filter method to estimate the airspeed with respect to body axis of an aircraft and wind speed together. It can also correct the scaling error of the measured airspeed. The proposed method uses GPS/INS navigation data and airspeed data. This method will be able to backup or replace the primary system to measure airflow angles. Estimated wind speed can also be used to reconstruct the GPS/INS navigation system by correcting the airspeed in the case of GPS failure.

2. AIRFLOW ANGLE AND WIND ESTIMATION

In many VTOL (vertical takeoff and landing) aircrafts, the Pitot-static tube of swivel-head type are used to measure airspeed (V_a) accurately in Fig 1.



Fig. 1 swivel-head airdata boom

The airspeed with respect to body axis, (u_a, v_a, w_a) are computed from true airspeed and airflow angle in equation (1).

$$\begin{aligned} u_a &= V_a \cos(\beta) \cos(\alpha) \\ v_a &= V_a \sin(\beta) \\ w_a &= V_a \cos(\beta) \sin(\alpha) \end{aligned} \quad (1)$$

The dynamic equation of motion of the rigid body aircraft is as follows.

$$\dot{\vec{V}}^I = \dot{\vec{V}}^B + \vec{\omega}^B \times \vec{V}^B = \vec{g} + \vec{F} / m \quad (2)$$

$\vec{\omega}$ and $\vec{g}$ represent the angular velocity of the aircraft and gravity vector. Upper index 'I' and 'B' mean inertial and body frame. The last term in equation (2), which includes aerodynamic and propulsive forces, is measured by accelerometer sensor of the navigation system.

Form the wind triangle of figure 2, the ground velocity is the sum of wind vector and the velocity vector with respect to the air.

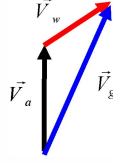


Fig. 2 wind triangle

$$\vec{V}_g = \vec{V}_a + \vec{V}_w \quad (3)$$

If the windy condition is constant, the equation (2) is applied to the velocity vector with respect to the air as well as the ground.

This paper proposes the Kalman filter algorithm to estimate airflow angles and wind vector. The airflow angles are computed by estimating the aircraft velocity with respect to the air as equation (1).

The state vector to be estimated by Kalman filter is defined in equation (4). The states are composed of body-axis components of airspeed vector and the wind vector components in navigation frame. The state vector also includes the scale factor to correct the error of the measured true airspeed. For example, this airspeed error is allowable by 3% in FAR (Federal Aviation Regulations) part 25.

$$\vec{x} = [u_a \quad v_a \quad w_a \quad sf \quad V_{w_N} \quad V_{w_E} \quad V_{w_D}]^T \quad (4)$$

The system dynamics of the Kalman filter are described by

$$\dot{\vec{x}} = F\vec{x} + \vec{u} + \vec{w} \quad (5)$$

The system matrix F is derived from equation (2). The scale factor and wind velocity components are modeled as random-walk processes. (a_x, a_y, a_z) are accelerometer measurements in equation (7).

$$F = \begin{bmatrix} 0 & r & -q & 0 & 0 & 0 & 0 \\ -r & 0 & p & 0 & 0 & 0 & 0 \\ q & -p & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (6)$$

$$\vec{u} = \begin{bmatrix} a_x - g \sin \theta \\ a_y + \sin \phi \cos \theta \\ a_z + g \cos \phi \cos \theta \end{bmatrix} \quad (7)$$

$$\vec{w} \sim N(0, Q) \quad (8)$$

Proposed Kalman filter uses true airspeed, ground velocity vector and lateral airspeed for measurement update of the filter.

The relation between measured true airspeed $\hat{V}_a$ and filter states is described by equation (9).

$$\hat{V}_a = h_1(\vec{x}) + v_1 = sf \cdot V_a = sf \cdot \sqrt{u_a^2 + v_a^2 + w_a^2} \quad (9)$$

The linearized observation matrix of the nonlinear equation (9) is as follows:

$$H_1 = \begin{bmatrix} \frac{\partial h_1}{\partial u_a} & \frac{\partial h_1}{\partial v_a} & \frac{\partial h_1}{\partial w_a} & \frac{\partial h_1}{\partial sf} & 0_{3 \times 3} \end{bmatrix} \quad (10)$$

$$= \begin{bmatrix} \frac{sf \cdot u_a}{V_a} & \frac{sf \cdot v_a}{V_a} & \frac{sf \cdot w_a}{V_a} & V_a & 0_{3 \times 3} \end{bmatrix}$$

$$v_1 \sim N(0, R_1) \quad (11)$$

The ground velocity is obtained from GPS/INS navigation system. From equation (3), the observation matrix of measured ground velocity components is defined by equation (12).

$$\hat{\vec{V}}_g = \begin{bmatrix} u_g \\ v_g \\ w_g \end{bmatrix} = \begin{bmatrix} I_{3 \times 3} & 0 & C_n^b \end{bmatrix} \vec{x} = H_2 \vec{x} \quad (12)$$

$$v_2 \sim N(0, R_2) \quad (13)$$

The lateral airspeed is also used for measurement update of the filter. The measured lateral velocity $\hat{v}_a$ is the product of measured true airspeed and side-slip angle. The side-slip angle can be estimated from navigation data and control input variables in the following section.

$$\hat{v}_a = h_3(\vec{x}) + v_3 = \hat{V}_a \sin(\beta) = sf \cdot v_a \quad (14)$$

$$H_3 = [0 \quad 1 \quad 0 \quad 1 \quad 0_{3 \times 3}] \quad (15)$$

$$v_3 \sim N(0, R_3) \quad (16)$$

3. SIDESLIP ANGLE COMPUTATION

The lateral dynamics of an aircraft extracted from equation (2) is:

$$\dot{v} + ur - wp = g \sin(\phi) \cos(\theta) + a_y \quad (17)$$

In the above equation, the lateral force on an aircraft is described by the sum of aerodynamic and propulsive force using stability and control derivatives. In the following equation, the lateral force includes the additional bias term related to aircraft asymmetry.

$$a_y \approx Y_v v + Y_p p + Y_r r + Y_\phi \phi + Y_{\delta_a} \delta_a + Y_{\delta_r} \delta_r + bias \quad (18)$$

In general, after these aerodynamic derivatives are

obtained from aircraft configuration and wind tunnel test, it is refined by flight data. In this paper, aerodynamic derivatives are computed from flight data using linear least-square method. The reference sideslip angle is measured by sideslip vane mechanically connected to Pitot-static tube. The control inputs δ_a and δ_r are known values from flight control system. Other data are obtained from GPS/INS navigation system.

$$\begin{bmatrix} a_y(1) \\ a_y(2) \\ \vdots \end{bmatrix} = \begin{bmatrix} v(1) & p(1) & r(1) & \phi(1) & \delta_a(1) & \delta_r(1) & 1 \\ v(2) & p(2) & r(2) & \phi(2) & \delta_a(2) & \delta_r(2) & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} Y_v \\ Y_r \\ Y_p \\ Y_\phi \\ Y_{\delta_a} \\ Y_{\delta_r} \\ b \end{bmatrix} \quad (Y = H \cdot X) \quad (19)$$

The solution by least-square method is:

$$X = (H^T H)^{-1} H^T Y \quad (20)$$

If the lateral velocity is approximated as the product of airspeed and sideslip angle, from equation (18), the sideslip angle is computed as equation (21).

$$\beta = \frac{a_y - Y_p p - Y_r r - Y_\phi \phi - Y_{\delta_a} \delta_a - Y_{\delta_r} \delta_r - bias}{Y_v \cdot V_a} \quad (21)$$

The above method needs accurate sideslip angle measurements as well as navigation data. During analyzing flight data in the early stage of flight test, the sideslip angle measured from mechanical vane was founded to have scaling error. Thus sideslip angle measurements were corrected before applying equation (17)~(21) [5].

To verify the proposed method, the real flight data of tilt-rotor UAV, Smart UAV were used which was developed by Korea Aerospace Research Institute. By linear least-square fit of eq. (20), computed stability and control derivatives are summarized in table 1.

Table 1 Stability and control derivatives derived by flight data

	80 KPH	100 KPH	120 KPH
Y_v	-0.000859	-0.000895	-0.001325
Y_r	0.0005849	-0.005051	-0.001514
Y_p	0.0111679	0.015141	0.016563
Y_ϕ	-0.002848	-0.003181	-0.001980
Y_{δ_a}	-0.130533	-0.057594	-0.105897
Y_{δ_r}	-0.036568	-0.074438	-0.065724
b	-0.018877	-0.031045	-0.060642

4. SIMULATION RESULTS

To verify proposed method, simulations were conducted using the Navion aircraft model. During simulations, the aircraft performed a 20° banking turn with the sinusoidal sideslip angle command of amplitude 6° under constant wind. The wind speed was set to 10 m/s and the wind direction was 90°. The airspeed was assumed to be measured 3% smaller than the true value.

Figure 3 shows true angle-of-attack and sideslip angle. The lower figure is sideslip angle measurements of simulation data. The standard deviation of sideslip angle measurements is 2°. The estimated airspeed and scale factor converged within 0.5 % of true values at about 30 sec in figure 4. The error of airspeed with respect to body axis was plotted with 3 sigma bound in figure 5. Figure 6 is the error plot of the angle-of-attack and side-slip angle computed from estimated airspeed. Figure 7 is the estimated wind speed. The result bounded by 3-sigma shows that the proposed filter works well.

The side-slip angle computation method used in this paper assumes flight regimes around level flight which side-slip angle is not too large. In figure 8, blue line is the same plots as figure 6 and red line is the result which doesn't use side-slip angle measurements after 100 second. Though airspeed measurements and constant wind condition were used, the accuracy of estimated airflow angles degraded only 0.1 degree. This result means that the proposed filter can be used under nonlinear condition if side-slip angle measurements are used or discarded appropriately.

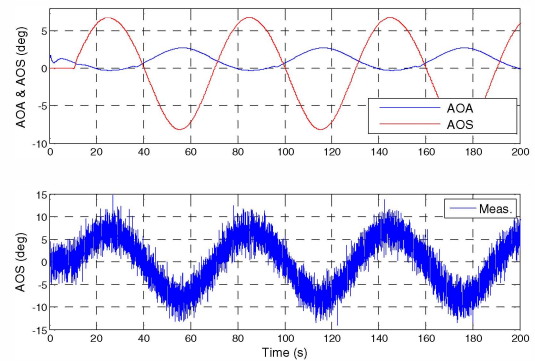


Fig. 3 lateral and down-word airspeed

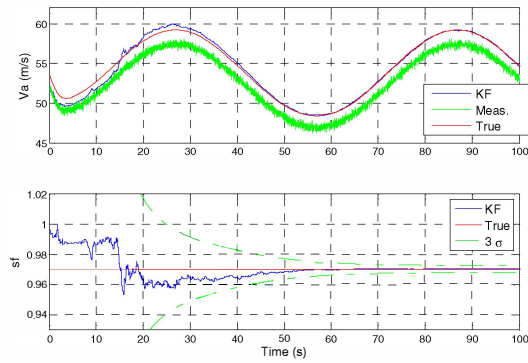


Fig. 4 Airspeed and scale factor

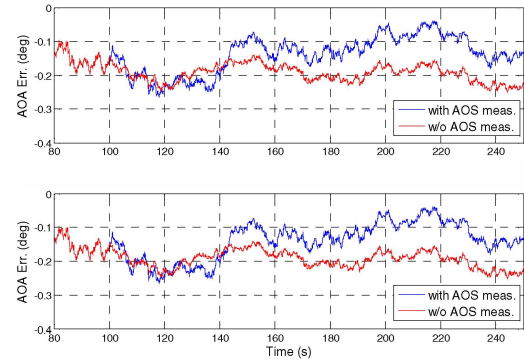


Fig. 8 The results with/without AOS measurements

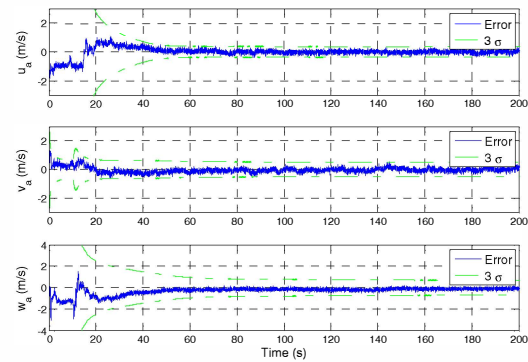


Fig. 5 Airspeed error

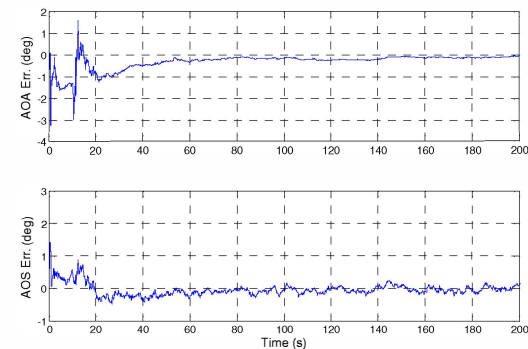


Fig. 6 AOA and AOS

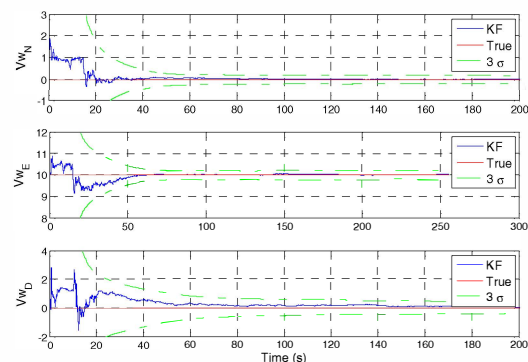


Fig. 7 Wind speed

5. CONCLUSION

In this paper, the Kalman filter based method was proposed to airflow angles of an aircraft and three-dimensional wind speed under constant wind condition. It also corrects measured airspeed. Proposed method uses airspeed measurements, constant wind condition and computed side-slip angle. The side-slip angle is computed using stability and control derivatives from flight data and navigation data. From simulation results, proposed method is expected to work well in flight regimes that sideslip angle can be computed from flight data. In the future, proposed method will be applied to real flight data.

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